CALCULUS FOR AI
Problem Set 01

From Slopes to Learning

A friendly introduction to derivatives, gradients, and how AI models learn

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The big idea
An AI model makes predictions. We measure how wrong those predictions are using a number called **loss**. Calculus helps us understand how that loss changes when we adjust the model. We can use this information to improve the model step by step.

Name: _____ **Date:** _____

Suggested time: 75–100 minutes **Total: 100 marks**

What you will learn

By the end of this problem set, you should be able to:
- describe how quickly something changes;
- understand a derivative as the slope at one point;
- find derivatives using a few basic rules;
- use the chain rule when functions are connected;
- find how a function changes when it has more than one input;
- explain how calculus helps an AI model reduce its loss.

A small guide to the symbols

You do not need to memorize these symbols immediately. Use this box whenever you need a reminder.

Symbols you will see

Symbol	Meaning in simple words
x or w	A number we can change. In an AI model, w can represent a setting the model learns.
$f(x)$	The output we get when we put x into a rule called f.
L	The loss: a number that measures how poorly the model is doing. Usually, lower is better.
$f'(x)$	The derivative: how quickly $f(x)$ is changing at the value x.
$\frac{dy}{dx}$	Another way to write the derivative of y with respect to x.
$\frac{\partial f}{\partial w_1}$	How f changes when w_1 changes while the other inputs stay fixed.
∇L (called "gradient of L")	A list of the derivatives for all the parameters. It shows the direction in which loss increases most quickly.
η (Greek letter "eta")	The learning rate: how large a step we take when updating a parameter.
$\lim_{h \to 0}$	Read this as "as h gets closer and closer to zero." It helps us describe change over a tiny interval.

How to work

- Work through the parts in order. Each part prepares you for the next.
- Show your steps. The way you think matters, not just the final answer.
- For explanation questions, use clear, everyday language.
- Include units when the question uses quantities such as kilometres or hours.
- You may use a calculator for arithmetic, but show how you set up the calculation.

Part A — Understanding change and slope (15 marks)

Start with a familiar question: how quickly is something changing?

A1. A bus travels 120 kilometres in 3 hours. **[3 marks]**

(a) Calculate its average speed.

(b) Can this information alone tell you the bus's exact speed 40 minutes after the trip began? Explain.

A2. A model's loss is 18 when its setting is $w = 2$, and its loss is 12 when its setting is $w = 5$. **[3 marks]**

(a) Find the average change in loss for each 1-unit increase in w between these settings.

(b) Is your answer positive or negative? Explain what that tells you.

A3. Suppose $f(x) = x^2$. We want to estimate the slope at $x = 2$. The table shows values just to the right of 2. **[4 marks]**

h	0.1	0.01	0.001	0.0001
$f(2+h)$	4.41	4.0401	4.004001	4.00040001

Here, h is the small distance we move to the right of $x = 2$, and $f(2) = 4$.

(a) For each value of h, calculate

$$\frac{f(2+h) - f(2)}{h}.$$

In words, this means: change in output divided by change in input.

(b) What number do your answers get closer to as h gets smaller?

(c) Why does this help us estimate the slope at exactly $x = 2$?

A4. What is the difference between a **secant line** and a **tangent line**? What does the slope of each line tell us? **[5 marks]**

Part B — Derivatives: measuring change at one point (20 marks)

A derivative tells us how steep a curve is at a particular point.

B1. We can find the derivative using this formula:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

It may look complicated, but it means: calculate the slope over a small interval, then imagine that interval getting closer to zero. For $f(x) = x^2$: **[5 marks]**

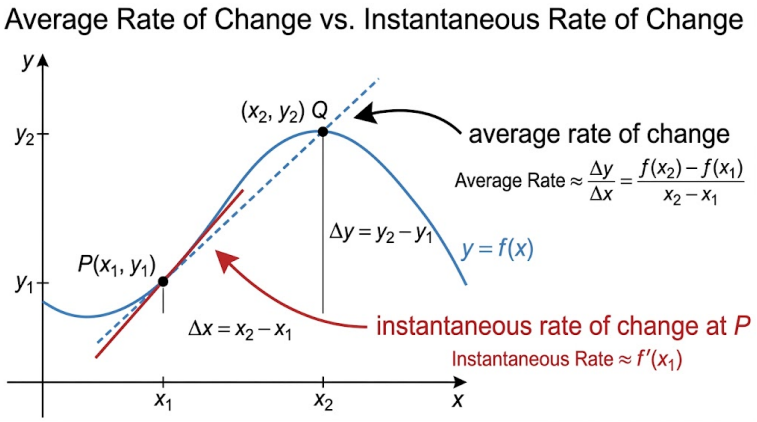


Figure 1: Average and instantaneous rate of change.

(a) Put x^2 in place of each $f(x)$ in the formula.
(b) Expand and simplify the fraction. Remember that $(x+h)^2 = x^2 + 2xh + h^2$.
(c) Let h get closer to zero and find the derivative $f'(x)$.

B2. Find the derivative of each function. **[6 marks]**

(a) $f(x) = 7$ (b) $g(x) = 5x^4$
(c) $h(x) = 3x^2 - 4x + 6$

Reminder: the derivative of a constant is 0. For a power such as x^n, bring the power down in front and reduce the power by 1.

B3. Let $p(x) = 2x^3 - 3x^2 + 4x - 1$. **[4 marks]**

(a) Find the derivative $p'(x)$.
(b) Find the slope when $x = 1$.
(c) At $x = 1$, is the function increasing or decreasing? Use your answer to explain.

B4. At the current setting of a model, the derivative of its loss is -4. **[3 marks]**

(a) If we increase the setting a little, do we expect the loss to rise or fall?
(b) If we want to reduce the loss, should we try increasing or decreasing the setting?

B5. True or false? Give a short reason for each answer. **[2 marks]**

(a) If the derivative is 0 at a point, that point must be the lowest point of the whole function.
(b) If the derivative is positive at a point, the function is rising there.

Part C — The chain rule: when changes pass through stages (15 marks)

In a neural network, one calculation feeds into the next. The chain rule helps us track how a change travels through these steps.

C1. Consider the function

$$y = (2x + 3)^3.$$

[5 marks]

(a) What is the inner expression? What is the outer operation?
(b) Use the chain rule to find the derivative $\dfrac{dy}{dx}$.
(c) Find the derivative when $x = 1$.

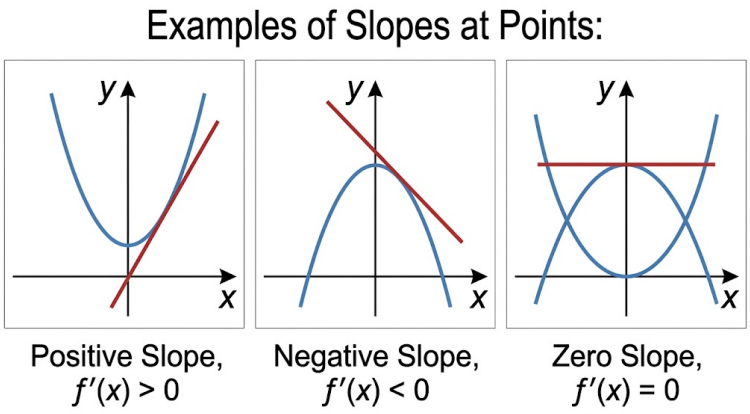


Figure 2: Positive, negative, and zero slopes.

C2. Imagine a small change passing through three stages of a model. Stage 1 multiplies the change by 2, stage 2 multiplies it by 3, and stage 3 multiplies it by 0.5. **[4 marks]**

(a) If the starting change is 0.1, what is the change at the output?
(b) Why do we multiply the three numbers? Explain how this connects to the chain rule.

C3. A student finds the derivative of $y = (x^2 + 1)^4$ and writes

$$\frac{dy}{dx} = 4(x^2 + 1)^3.$$

What part did the student forget? Write the correct derivative. **[3 marks]**

C4. An early layer in a neural network affects later layers, and those layers affect the final loss. In your own words, explain how the chain rule helps us work out the effect of changing an early layer. **[3 marks]**

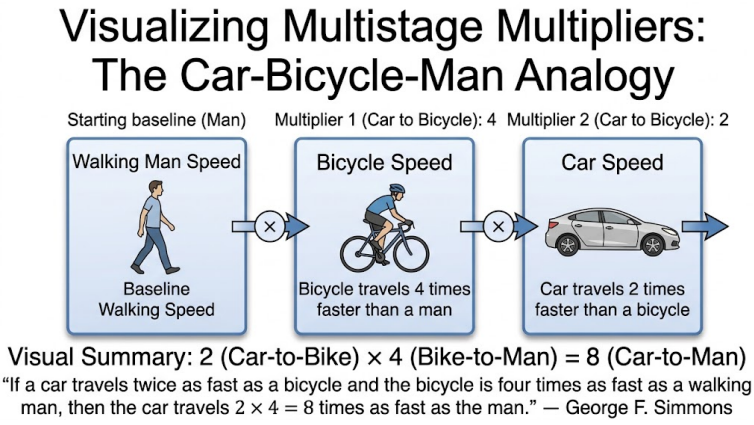


Figure 3: A change passing through three connected stages.

Part D — When there is more than one setting (20 marks)

A model may have many settings. A partial derivative looks at one setting at a time. The gradient puts those results together.

A quick reminder

The symbol $\frac{\partial f}{\partial w_1}$ means: measure how f changes when w_1 changes, while keeping the other input fixed. The gradient is the list of these rates of change for all the settings.

D1. Consider

$$f(w_1, w_2) = 3w_1^2 + 4w_1w_2 + w_2^3.$$

Here, w_1 and w_2 are two numbers we can change. **[7 marks]**

(a) Find how f changes with w_1, keeping w_2 fixed.
(b) Find how f changes with w_2, keeping w_1 fixed.
(c) Calculate both answers when $w_1 = 1$ and $w_2 = 2$.

D2. In Question D1, why do we keep w_2 fixed when finding how f changes with w_1? Explain what changes and what stays the same. **[3 marks]**

D3. The gradient is a list of the partial derivatives. It points in the direction where the loss rises most quickly. Suppose the gradient is

$$\nabla L = [-6,\ 2].$$

The symbol ∇L is read as "gradient of L." **[4 marks]**

(a) Does the gradient point toward increasing loss or decreasing loss?
(b) Write the negative gradient by changing the sign of each number.
(c) If we take a small step in the negative-gradient direction, should w_1 increase or decrease? What about w_2?

D4. A simple model has the loss

$$L(w_1, w_2) = (w_1 - 2)^2 + (w_2 + 1)^2.$$

[6 marks]

(a) Find the gradient.
(b) Find the gradient when $w_1 = 0$ and $w_2 = 0$.
(c) Use the negative gradient to say which way each setting should move to reduce the loss locally.

Part E — A complete learning step (20 marks)

Now use the ideas together to see how a model can adjust one setting to reduce its loss.

A simple model has one setting, w. Its loss is

$$L(w) = (w - 3)^2 + 1.$$

Our goal is to choose a value of w that makes the loss as small as possible.

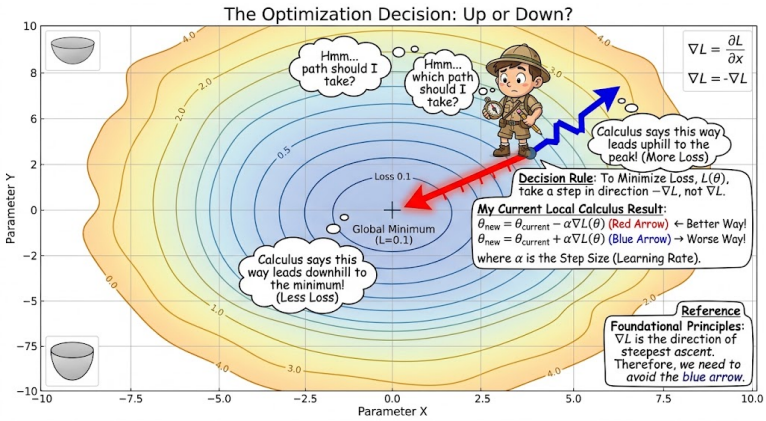


Figure 4: Understanding the gradient and the direction of decreasing loss.

E1. Find the derivative of the loss, $L'(w)$. **[3 marks]**

E2. When $w = 1$, calculate the loss $L(1)$ and the derivative $L'(1)$. **[3 marks]**

E3. We update the setting using this rule:

$$\text{new setting} = \text{old setting} - (\text{learning rate} \times \text{derivative}).$$

In symbols:

$$w_{\text{new}} = w_{\text{old}} - \eta L'(w_{\text{old}}).$$

The learning rate, η, controls the size of the step. Start with $w = 1$ and use a learning rate of 0.1. Find the new setting. Show your calculation. **[4 marks]**

E4. Calculate the loss before and after this update. Did the loss go down? Use the numbers to explain. **[4 marks]**

E5. Start again from $w = 1$, but now use a learning rate of 1.2. Find the new setting and its loss. Compare this with the starting loss and your answer to E4. What does this show about choosing the learning rate? **[4 marks]**

E6. What value of w gives the smallest possible loss? What is that loss? Explain briefly. **[2 marks]**

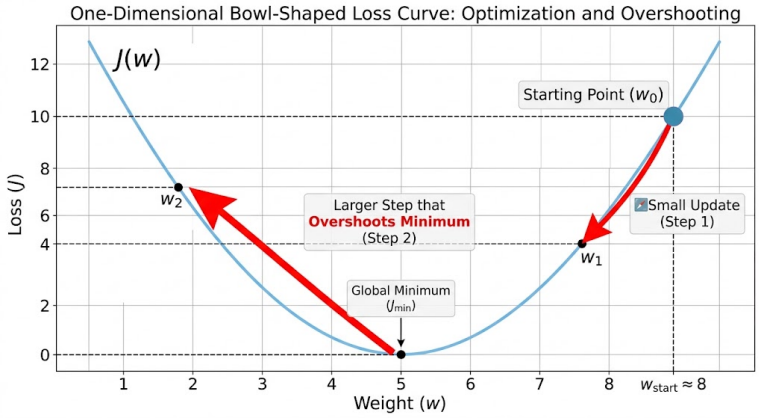


Figure 5: A bowl-shaped loss curve and parameter updates.

Part F — Explain the big idea (10 marks)

These questions are about explaining the meaning, not just doing calculations.

F1. A classmate says, "The gradient tells the AI the correct answer." Why is this not quite right? What does the gradient tell us instead? **[3 marks]**

F2. Suppose the derivative of the loss with respect to a setting is positive. Explain why subtracting a small positive multiple of that derivative may reduce the loss. **[3 marks]**

F3. Give an everyday example of something that changes. State:

(a) what the input is;
(b) what quantity depends on that input;
(c) what the instantaneous rate of change means in your example.

[4 marks]

The goal is not just to calculate a derivative. It is to understand what change means.